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Predicting Chronic Kidney Disease (CKD) Using Biomarkers

Approximately 850 million persons worldwide are estimated to have kidney disease, most of whom live in low-income and lower-middle-income countries (LICs and LMICs). A large proportion of these individuals do not have access to kidney disease diagnosis, prevention or treatment. Up to 9 out of 10 individuals with CKD in resource-poor settings with weak primary care infrastructure are unaware that they have this condition and therefore do not seek treatment(Anna *et al.,*2024).

Currently, kidney disease is the third fastest-growing cause of death globally and the only non-communicable disease to exhibit a continued rise in age-adjusted mortality. By 2040, CKD is projected to be the 5th highest cause of years of life lost (YLL) globally(Anna *et al.,*2024).

About 1 in 3 people with diabetes and 1 in 5 people with high blood pressure have kidney disease. Other risk factors for developing kidney disease include heart disease and a family history of kidney failure. Despite the prevalence of kidney disease in the United States, as many as 9 in 10 adults who have CKD are unaware they have the disease. Early-stage kidney disease usually has no symptoms, and many people don't know they have CKD until it is very advanced (NIDDK).

Biomarkers are measurable indicators that provide valuable insights into biological processes, pathogenic mechanisms, and responses to therapeutic interventions. In CKD, biomarkers are crucial for various aspects of patient care. Traditional biomarkers for CKD include serum creatinine, blood urea nitrogen (BUN), and urine protein and albumin. Serum creatinine is widely used to estimate kidney function and calculate the estimated GFR (eGFR)(Aman *et l.,* 2024).

Early-stage CKD doesn't usually cause symptoms, but your doctor may recommend an eGFR test if you are at higher risk of developing the disease. Glomerular filtration rate (GFR) is the best overall index for assessing kidney function (National Kidney Foundation).

About The Dataset

The dataset is synthetic from Kaggle.
CKD Dataset

Overview of the Dataset

This dataset contains detailed health information for 1,659 patients diagnosed with Chronic Kidney Disease (CKD). The dataset includes demographic details, lifestyle factors, medical history, clinical measurements, medication usage, symptoms, quality of life scores, environmental exposures, and health behaviours. Each patient is uniquely identified by a Patient ID, and the data includes a confidential column indicating the doctor in char

1. **Patient Information**
    - Patient ID
    - Demographic Details
    - Lifestyle Factors

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7. **Health Behaviors**
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Patient Information

- **PatientID**: A unique identifier assigned to each patient (1 to 1,659).

Demographic Details

- **Age**: The age of the patients ranges from 20 to 90 years.
- **Gender**: Gender of the patients, where 0 represents Male and 1 represents Female.
- **Ethnicity**: The ethnicity of the patients, coded as follows:
  - 0: Caucasian
  - 1: African American
  - 2: Asian
  - 3: Other
- **SocioeconomicStatus**: The socioeconomic status of the patients, coded as follows:
  - 0: Low
  - 1: Middle
  - 2: High
- **EducationLevel**: The education level of the patients, coded as follows:
  - 0: None
  - 1: High School
  - 2: Bachelor's
  - 3: Higher

Lifestyle Factors

- **BMI**: Body Mass Index of the patients, ranging from 15 to 40.
- **Smoking**: Smoking status, where 0 indicates No and 1 indicates Yes.
- **AlcoholConsumption**: Weekly alcohol consumption in units, ranging from 0 to 20.
- **PhysicalActivity**: Weekly physical activity in hours, 0 to 10.

- **DietQuality**: Diet quality score, ranging from 0 to 10.
- **SleepQuality**: Sleep quality score, 4 to 10.

Medical History

- **FamilyHistoryKidneyDisease**: Family history of kidney disease, where 0 indicates No and 1 indicates Yes.
- **FamilyHistoryHypertension**: Family history of hypertension, where 0 indicates No and 1 indicates Yes.
- **FamilyHistoryDiabetes**: Family history of diabetes, where 0 indicates No and 1 indicates Yes.
- **PreviousAcuteKidneyInjury**: History of previous acute kidney injury, where 0 indicates No and 1 indicates Yes.
- **UrinaryTractInfections**: History of urinary tract infections, where 0 indicates No and 1 indicates Yes.

Clinical Measurements

- **SystolicBP**: Systolic blood pressure, 90 to 180 mmHg.
- **DiastolicBP**: Diastolic blood pressure, 60 to 120 mmHg.
- **FastingBloodSugar**: Fasting blood sugar levels from 70 to 200 mg/dL.
- **HbA1c**: Hemoglobin A1c levels, 4.0% to 10.0%.
- **SerumCreatinine**: Serum creatinine levels, 0.5 to 5.0 mg/dL.
- **BUNLevels**: Blood Urea Nitrogen levels, ranging from 5 to 50 mg/dL.
- **GFR**: Glomerular Filtration Rate, ranging from 15 to 120 mL/min/1.73 m^2.
- **ProteinInUrine**: Protein levels in urine, 0 to 5 g/day.
- **ACR**: Albumin-to-Creatinine Ratio, 0 to 300 mg/g.
- **Serum Electrolytes Sodium**: Serum sodium levels, 135 to 145 mEq/L.
- **SerumElectrolytesPotassium**: Serum potassium levels, 3.5 to 5.5 mEq/L.
- **SerumElectrolytesCalcium**: Serum calcium levels, 8.5 to 10.5 mg/dL.
- **SerumElectrolytesPhosphorus**: Serum phosphorus levels, 2.5 to 4.5 mg/dL.
- **HemoglobinLevels**: Hemoglobin levels, 10 to 18 g/dL.
- **CholesterolTotal**: Total cholesterol levels, 150 to 300 mg/dL.
- **CholesterolLDL**: Low-density lipoprotein cholesterol levels, 50 to 200 mg/dL.
- **CholesterolHDL**: High-density lipoprotein cholesterol, 20 to 100 mg/dL.
- **CholesterolTriglycerides**: Triglycerides levels, 50 to 400 mg/dL.

Medications

- **ACEInhibitors**: Use of ACE inhibitors, where 0 indicates No and 1 indicates Yes.
- **Diuretics**: Use of diuretics, where 0 indicates No and 1 indicates Yes.
- **NSAIDsUse**: Frequency of NSAIDs use, ranging from 0 to 10 times per week.
- **Statins**: Use of statins, where 0 indicates No and 1 indicates Yes.
- **AntidiabeticMedications**: Use of antidiabetic medications, where 0 indicates No and 1 indicates Yes.

Symptoms and Quality of Life

- **Edema**: Presence of edema, where 0 indicates No and 1 indicates Yes.
- **FatigueLevels**: Fatigue levels, 0 to 10.
- **NauseaVomiting**: Frequency of nausea and vomiting, ranging from 0 to 7 times per week.
- **MuscleCramps**: Frequency of muscle cramps, ranging from 0 to 7 times per week.
- **Itching**: Itching severity, ranging from 0 to 10.
- **QualityOfLifeScore**: Quality of life score, 0 to 100.

Environmental and Occupational Exposures

- **HeavyMetalsExposure**: Exposure to heavy metals, where 0 indicates No and 1 indicates Yes.
- **OccupationalExposureChemicals**: Occupational exposure to harmful chemicals, where 0 indicates No and 1 indicates Yes.
- **WaterQuality**: Quality of water, where 0 indicates Good and 1 indicates Poor.

Health Behaviors

- **MedicalCheckupsFrequency**: Frequency of medical check-ups per year, ranging from 0 to 4.
- **MedicationAdherence**: Medication adherence score, 0 to 10.
- **HealthLiteracy**: Health literacy score, 0 to 10.

Diagnosis Information

- **Diagnosis**: Diagnosis status for Chronic Kidney Disease, where 0 indicates No and 1 indicates Yes.

Confidential Information

- **DoctorInCharge**: This column contains confidential information about the doctor in charge, with "Confidential" as the value for all patients.
  This comprehensive dataset provides valuable insights into the various factors associated with Chronic Kidney Disease and can be used for various analyses, including statistical analysis, machine learning model development, and more.

TABLE 1: Clinical Measurements and Their Significance to CKD

Measurement	Meaning & Physiological Significance	Normal Range	High Level & Significance	Low Level & Significance	Path to CKD
GFR	**Filtration Rate:** The volume of fluid filtered by the kidney per minute.	90–120 mL/min	**Hyperfiltration:** Often seen in early diabetes; stresses the filters.	**Kidney Failure:** Toxins and fluids accumulate dangerously in the body.	Sustained low GFR is the primary diagnostic criterion for CKD.
Serum Creatinine	**Muscle Waste:** A byproduct of muscle metabolism cleared solely	0.6–1.3 mg/dL	**Renal Impairment:** Indicates the "drain" is clogged; waste is backing up.	**Muscle Wasting:** Can indicate frailty or advanced	Rising levels track the loss of functioning nephrons

Measurement	Meaning & Physiological Significance	Normal Range	High Level & Significance	Low Level & Significance	Path to CKD
	by the kidneys.			malnutrition.	(kidney units).
BUN	**Protein Waste:** Measures nitrogen from urea (protein byproduct).	7–20 mg/dL	**Uremia:** High nitrogen causes nausea, mental fog, and "uremic frost."	**Liver/Nutrition Issues:** Suggests low protein intake or liver dysfunction.	Chronic high urea causes systemic toxicity and cardiovascular stress.
Protein in Urine	**Structural Marker:** Measures large proteins leaking through the filters.	0–150 mg/day	**Glomerular Damage:** The physical "mesh" of the kidney filter is torn.	**Healthy State:** Zero or trace amounts are the target for health.	Leaking protein is inflammatory and physically scars the kidney tubules.
ACR (Albumin / Creatinine Ratio)	**Early Leakage:** A sensitive test for tiny amounts of albumin	< 30 mg/g	**Micro/Macroalbuminuria:** Earliest sign of diabetic kidney damage.	**Healthy State:** Indicates the kidney's blood-urine barrier is intact.	High ACR predicts future GFR decline, especially

Measurement	Meaning & Physiological Significance	Normal Range	High Level & Significance	Low Level & Significance	Path to CKD
	(microalbuminuria).				in diabetics.
Hemoglobin	**Oxygen Carrier:** Kidneys produce EPO to signal red blood cell production.	12–18 g/dL	**Polycythemia:** Often due to dehydration or smoking in this context.	**Anemia:** The kidney can't produce EPO. Leads to extreme fatigue.	Anemia deprives the remaining kidney cells of oxygen, killing them faster.
Fasting Blood Sugar	**Glucose Level:** Amount of sugar in the blood after fasting.	70–100 mg/dL	**Hyperglycemia:** High sugar "caramelizes" and scars small blood vessels.	**Hypoglycemia:** Dangerous drop in energy; can lead to seizures or coma.	**Diabetes** (High Sugar) is the #1 global cause of kidney destruction.

Measurement	**Meaning & Physiological Significance**	**Normal Range**	**High Level & Significance**	**Low Level & Significance**	**Path to CKD**
HbA1c	**Average Glucose:** 3-month average blood sugar control.	4.0%–5.6%	**Chronic Diabetes:** Indicates long-term "wear and tear" on kidney filters.	**Anemia/Blood Loss:** Can be falsely low if red blood cells die too fast.	High HbA1c leads to irreversible scarring of the glomerular basement membrane.
Systolic BP	**Heart Force:** Pressure during a heartbeat.	90–120 mm Hg	**Hypertension:** Physically hammers and bursts delicate kidney capillaries.	**Hypoperfusion:** The kidney isn't getting enough blood to function.	High pressure "burns out" the kidney's internal plumbing over time.
Diastolic BP	**Resting Pressure:** Pressure between heartbeats.	60–80 mm Hg	**Vascular Strain:** High resting pressure damages the renal arteries.	**Low Flow:** Can lead to ischemic injury (cell death from low blood).	Consistent high pressure is the #2 cause of CKD after diabetes.

Measurement	Meaning & Physiological Significance	Normal Range	High Level & Significance	Low Level & Significance	Path to CKD
Sodium	**Water Balance:** Controls fluid distribution between cells and blood.	135–145 mEq/L	**Hypernatremia:** Causes thirst, high BP, and fluid retention (Edema).	**Hyponatremia:** Causes brain cell swelling, confusion, and seizures.	Poor sodium handling leads to the hypertension that drives CKD.
Potassium	**Heart/Nerve Signal:** Vital for the electrical rhythm of the heart.	3.5–5.5 mEq/L	**Hyperkalemia: Fatal Risk;** can cause sudden cardiac arrest.	**Hypokalemia:** Causes muscle paralysis and irregular heartbeats.	As CKD worsens, kidneys lose the ability to flush life-threatening potassium.
Calcium	**Mineral Signal:** Essential for bones, nerves, and heart muscle.	8.5–10.5 mg/dL	**Stones/Calcification:** Can lead to kidney stones and vessel hardening.	**Hypocalcemia:** Causes spasms and triggers bone-wasting hormones.	Low calcium in CKD leads to "Renal Bone Disease" (brittle bones).

Measurement	Meaning & Physiological Significance	Normal Range	High Level & Significance	Low Level & Significance	Path to CKD
Phosphorus	**Mineral Balance:** Works with calcium; cleared by healthy kidneys.	2.5–4.5 mg/dL	**Hyperphosphatemia:** Binds with calcium to form "bone" in soft tissues.	**Muscle Weakness:** Rare in CKD; usually due to severe malnutrition.	High phosphorus calcifies (turns to stone) the heart and renal arteries.
Total Cholesterol	**Blood Fats:** Sum of all cholesterol types in the blood.	150–200 mg/dL	**Atherosclerosis:** Clogs the renal arteries, starving the kidney.	**Malnutrition:** Can indicate severe systemic illness in late-stage CKD.	High fats cause "Lipotoxicity," where fat deposits kill kidney cells.
LDL Cholesterol	**"Bad" Fat:** Deposits cholesterol into artery walls.	50–100 mg/dL	**Plaque Buildup:** Directly narrows the vessels feeding the kidneys.	**Low Hormone Prod:** Rare concern; usually seen as very healthy.	Narrows the renal artery, causing "Renal Artery Stenosis" (starved kidney).

Measurement	Meaning & Physiological Significance	Normal Range	High Level & Significance	Low Level & Significance	Path to CKD
HDL Cholesterol	**"Good" Fat:** Scavenges bad fats for disposal.	40–60 mg/dL	**Protective:** High levels help prevent vascular damage in kidneys.	**High Cardiac Risk:** Increases risk of heart attack, common in CKD.	Low HDL allows plaque to build up faster in the kidney's blood supply.
Triglycerides	**Energy Fats:** Fats stored for energy; found in the blood.	50–150 mg/dL	**Inflammation:** High levels are toxic to the kidney's filtration cells.	**Low Energy/Fats:** Can be caused by malabsorption or strict diet.	Contributes to the "metabolic syndrome" that accelerates renal decline.

TABLE 2: Some Medications in the Dataset and how they affect CKD

Medication	Who takes it? (Target Patients)	What it does in the body	Impact on CKD

ACE Inhibitors	Patients with **Hypertension** (High BP), **Diabetes**, or high **Protein in Urine**.	Relaxes blood vessels by blocking the hormone Angiotensin II; specifically dilates the "exit" valve of the kidney's filters.	**Protective:** Lowers the internal pressure that physically tears the kidney's "sieve," stopping protein leakage.
Diuretics	Patients with **Edema** (swelling), fluid retention, or resistant **High Blood Pressure**.	Prompts the kidneys to flush out more Sodium into the urine, which pulls excess water out of the blood.	**Symptom Control:** Reduces the "workload" of the heart and kidneys by decreasing total blood volume and swelling.
Insulin	Patients with **Diabetes** (High Fasting Blood Sugar or HbA1c > 6.5%).	Acts as a "key" to let sugar leave the bloodstream and enter cells to be used for energy.	**Prevention:** Stops high sugar from "caramelizing" and hardening the delicate capillaries in the kidney.
Statins	Patients with high **LDL** or **Total Cholesterol** (Dyslipidemia).	Blocks the enzyme in the liver that produces cholesterol and helps stabilize artery walls.	**Vascular Health:** Keeps the **Renal Artery** open, ensuring the kidney continues to receive oxygen-rich blood.

NSAIDs	People seeking relief for **Pain or Inflammation** (though often restricted in CKD).	Blocks "Prostaglandin" chemicals that normally keep blood vessels open.	**Harmful:** For CKD patients, these "kink the hose," starving the kidney of blood flow and causing sudden drops in GFR.

Risk Factors of CKD

Anyone can develop CKD - at any age. However, some people are more at risk than others. The most common CKD risk factors include:

- Diabetes
- Personal history of acute kidney injury (AKI)
- Obesity
- Over the age of 60
- Smoking and/or use of tobacco products
- High blood pressure (hypertension)
- Heart disease and/or heart failure
- Family history of CKD or kidney failure
- urinary tract infections (UTIs)

(National Kidney Foundation)

GFR

Glomerular filtration rate (GFR) is the best overall kidney function index. Normal GFR varies by gender, age and body size, and declines with age (National Kidney Foundation).

- eGFR of 90 or higher is in the normal range
- eGFR of 60 -89 may mean early-stage kidney disease
- eGFR of 15 -59 may mean kidney disease

- eGFR below 15 may mean kidney failure (National Kidney Foundation)

In adults, the normal eGFR number is usually over 90. eGFR decreases with age, even in people without kidney disease. See the chart below for average estimated eGFR based on age(National Kidney Foundation).

Table 3: Average eGFR with Age Range

Age (years)	Average eGFR
20–29	116
30–39	107
40–49	99
50–59	93
60–69	85
70+	75

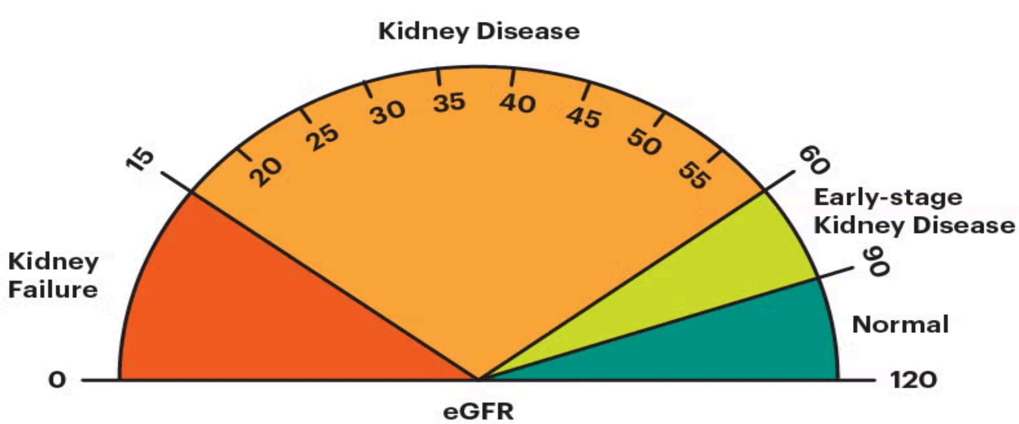

Fig 1: Physiological Range of eGFR(estimated GFR)(National Kidney Foundation)

Protein in Urine (Proteinuria) as a Driver of CKD

Proteinuria is not just a sign that the kidney's filters (glomeruli) are leaking; it is "nephrotoxic," meaning the presence of protein where it doesn't belong actively destroys kidney tissue over time. In a healthy kidney, the glomerular filtration barrier prevents large proteins like albumin from passing into the urine. When this barrier is damaged (by diabetes or hypertension), protein leaks into the renal tubules. The cells lining the kidney tubules try to reabsorb this leaking protein. This "overwork" triggers a pro-inflammatory response, leading to **tubulointerstitial inflammation** and **fibrosis** (scarring). This scarring eventually destroys the nephrons, leading to the progression of CKD (Cravedi & Remuzzi, 2013).

BUN (Blood Urea Nitrogen) as an Indicator of CKD

Urea is the primary metabolite derived from dietary protein and tissue protein breakdown. It is filtered at the glomerulus freely but not secreted, and it is reabsorbed by the renal tubules. Additionally, as urine flow rates decrease, more urea is reabsorbed. Blood urea nitrogen (BUN) measures level of nitrogen in the serum urea. BUN levels are inversely correlated with the decline of kidney function and are also affected by extrarenal factors like protein intake, gastrointestinal bleeding, catabolic states, malnutrition, heart failure, dehydration, use of glucocorticoids, and hepatic urea synthesis (Seki *et al.*, 2019)

High BUN does not typically *cause* CKD; rather, it is a metabolic waste product that accumulates because the kidneys are no longer able to filter it out. Urea is a byproduct of protein breakdown in the liver. In healthy kidneys, urea is filtered from the blood and excreted. As CKD progresses and the **Glomerular Filtration Rate (GFR)** drops, urea stays in the bloodstream, raising BUN levels. While high BUN is an indicator, the "uremic toxins" that rise alongside it can cause systemic symptoms like fatigue, nausea, and "brain fog," further complicating the health of a CKD patient.

Conclusion

Chronic Kidney Disease (CKD) is a significant global health challenge, currently ranked as the third fastest-growing cause of death and projected to be a leading cause of years of life lost by 2040. The disease is particularly dangerous because it is often asymptomatic in its early stages, leaving up to 90% of affected individuals unaware of their condition until it reaches an advanced phase. Major risk factors include diabetes and hypertension, which damage the kidney's delicate filtering units and lead to a progressive decline in the Glomerular Filtration Rate (GFR).

The progression of the disease is closely tied to the behavior of specific biomarkers, most notably proteinuria and Blood Urea Nitrogen (BUN). Proteinuria occurs when the glomerular barrier fails, allowing proteins to leak into the urine; this process is nephrotoxic, as the tubules' attempt to reabsorb these proteins triggers inflammation and permanent scarring (fibrosis). In contrast, BUN serves as a critical indicator of filtration capacity. While urea itself is a metabolic byproduct of protein breakdown, its accumulation in the bloodstream signals that the kidneys are no longer effectively clearing waste, leading to systemic "uremic" symptoms like nausea and cognitive impairment.

Effective CKD management relies on early detection and the stabilization of these biological markers to prevent total kidney failure. Diagnostic tools like the Urine Albumin-to-Creatinine Ratio (UACR) and blood tests for creatinine and BUN enable clinicians to monitor renal health and intervene with treatments such as ACE inhibitors. By managing underlying conditions and reducing the inflammatory load caused by protein leakage, it is possible to slow the transition from early-stage damage to irreversible renal scarring, ultimately improving long-term patient outcomes and quality of life.

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